Vlad-Emil Flore

Software Engineer (Java | Python | GenAI)

DETAILS

flore.vlad@gmail.com

Barcelona

Romania

EDUCATION

Master of Science in Informatics
West University of Timișoara
2008 — 2010
Focus on Artificial Intelligence and Distributed Computing
Grade: 9/10

Bachelor of Science in Informatics
West University of Timișoara
2004 — 2008
Grade: 9.67/10

LANGUAGES

- Romanian (native)
- English (C1)
- German (B2+)
- Spanish (A2)

PROFILE

- Backend Software Engineer specializing in Java and Python, building scalable, cloud-native applications and modernizing complex legacy systems
- Versatile project experience including greenfield and brownfield initiatives in public sector services, content management systems, and intelligence document processing solutions
- Proven leader in agile development environments, driving team collaboration, ensuring code quality, and meeting project deadlines

EMPLOYMENT HISTORY

Independent Software Engineer

Freelancing, Remote (Barcelona)
January 2025 - present

- Designed, developed, and maintained Web APIs (Java and Python), including integration with external services
- Created educational software engineering content focused on Java and the JVM ecosystem, Data Structures and Algorithms
- Designed and implemented LLM- and RAG-based systems for document understanding and question answering
- Built end-to-end machine learning pipelines from data preprocessing to model deployment

SKILLS

- Programming Languages (Java, Python)
- Machine Learning and Generative AI
- Testing (JUnit, Mockito, AssertJ, Pytest, Testcontainers)
- Coding Practices (Clean Code, TDD, SOLID)
- Web API (RESTful APIs, GraphQL, OpenAPI, Swagger, PostGraphile)
- Web Frameworks (Spring Framework, Spring Boot, Django, FastAPI)
- Microservices & Cloud-Native Architecture
- Messaging and Event-Sourcing Systems (RabbitMQ, EventStoreDB)
- Databases & ORM (PostgreSQL, Oracle, (Spring Data) JPA/Hibernate, SQLAlchemy)
- Cloud (AWS, Kubernetes, Keda, Docker)
- Build Tools (Maven, Gradle, Bazel)
- CI/CD (Git, GitHub, GitHub Actions, GitLab, TeamCity, Argo CD, Jenkins, Helm, Ansible)
- Team Leadership, Mentoring & Knowledge Sharing

> **Tech Stack:** Java, Spring (Framework, Boot, AI), Python, FastAPI, Django, PyScript, OpenAPI, Weaviate, PostgreSQL (with pgvector), Generative AI (LLM, RAG, Agentic Systems), LangChain(4j), Ollama, Docker, Kubernetes, AWS (EC2, S3, IAM, Lambda, ECR, ECS, EKS, Bedrock, SageMaker)

Educational Content Creator

Contract, Hyperskill, Remote (Barcelona)

October 2023 — March 2024

- Created and enhanced educational content for the Spring Boot track offered by JetBrains Academy through the Hyperskill platform, contributing to high-quality, hands-on learning experiences for aspiring developers

> **Tech Stack:** Java, Spring Framework, Spring Boot

Software Engineer

Employment, ExB Group, Remote (Nürnberg, Barcelona)

March 2021 — November 2024

- Implemented key features in Java and Python within a microservices-based event-driven architecture for a web-based Intelligent Document Processing application
- Designed and developed scalable, decoupled microservices in Java and Python, enhancing system robustness and performance (documents upload, archiving and download, image processing, REST and GraphQL APIs, notifications, AI models deployment)
- Deployed services in a cloud-native environment using Docker and Kubernetes, ensuring high availability and scalability
- Led internal technical workshops focused on clean code principles and improving overall code quality
- Conducted regular code reviews and participated in pair programming to foster collaboration and knowledge sharing
- Advocated for tooling and workflow improvements to streamline the developer experience and reduce friction in daily tasks

LINKS

- Certifications
- LinkedIn
- GitHub

> **Tech Stack:** Python, Java, Django, Spring Framework, Spring Boot, Akka, GraphQL, PostGraphile, PostgreSQL, EventStoreDB, RabbitMQ, Docker, Kubernetes, Helm, Keda, AWS S3, MinIO, AWS ECR, AWS ECS, AWS EKS, Git, GitLab, TeamCity, Argo CD, Grafana, AsciiDoc, Antora, Linux, JUnit

Lead Software Developer

Contract, IT-Systemhaus der Bundesagentur für Arbeit, on-site (Nürnberg)

September 2019 — April 2020

- Served as Lead Developer for one of two agile development teams, guiding technical direction and team collaboration
- Defined and evolved the application's architecture to support scalability and maintainability
- Promoted clean code and consistent development practices to maintain code quality
- Collaboratively decomposed features into manageable tasks, ensuring effective sprint planning and delivery
- Implemented and tested application features across the full stack, maintaining close alignment with business requirements

> **Tech Stack:** Java SE 8, Java EE 7, Angular, TypeScript, JavaScript, JUnit, Cucumber, Jenkins, Docker, Linux, Oracle Database, Git

Full Stack Developer

Contract, Bundesamt für Migration und Flüchtlinge (BAMF), on-site (Nürnberg)

May 2018 — September 2019

- Implemented features across both frontend and backend of a web application
- Improved and maintained project documentation to support developer onboarding and knowledge sharing
- Managed DevOps-related tasks and supported deployment activities to ensure smooth delivery and system stability

> **Tech Stack:** Java SE 8, Spring Framework, Spring Boot, JUnit, Angular, TypeScript, JavaScript, Ansible, Jenkins, Docker, Oracle Database, Hibernate, Linux, Git

Software Developer

Contract, IT-Systemhaus der Bundesagentur für Arbeit, on-site (Nürnberg)

May 2014 — February 2018

- Developed a publishing workflow for web content within Oracle CMS, streamlining the content creation and approval process
- Extended the CMS by implementing custom tools and features to support efficient content management and improve the user experience for content editors

> **Tech Stack:** Oracle WebCenter Content, Java, HTML, CSS, JavaScript, Oracle Database, Linux, CVS, Git

Senior Technical Consultant

Employment, virtual7 GmbH, on-site (Nürnberg)

February 2019 — May 2020

- Delivered software solutions across various client projects as part of agile development teams
- Implemented new features, resolved bugs, and participated in the full SDLC
- Acted as a software consultant, adapting quickly to different domains and tech stacks

Software Developer / Senior Technical Consultant

Employment, virtual7 Romania, on-site (Timișoara)

February 2008 — January 2019

- Delivered software solutions across various client projects as part of agile development teams
- Implemented new features, resolved bugs, and participated in the full SDLC across various client engagements
- Acted as a software consultant, adapting quickly to different domains and tech stacks